

AST1410 MESA Practical 2

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For consistency (and because my file organization was very bad for the first practical), the original simulation was rerun up to the post MS turnoff criterion of minimum core H1 fraction 10^{-3} alongside the two new stars with `mixing_length_alpha` = 1, 4. The evolutionary trajectories of all three stars are shown in the same plots below. The main modifications made to the files from last time were: the removal of `inlist_load`; the reinstatement of the stopping condition mentioned above; the specification of the `mixing_length_alpha` parameter; additions to `plot.py`, which is used to generate all the plots. All files are uploaded to the main branch on https://github.com/Gursach44/MESA_practicals/tree/main/MESA_practical.2.

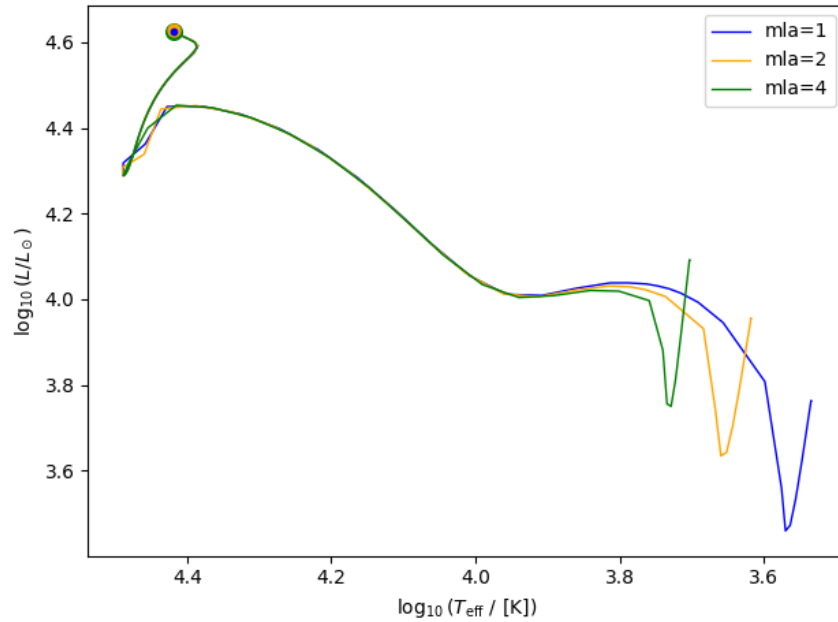


Figure 1: HR diagram showing evolution of 15 solar mass stars with varying `mixing_length_alpha` parameter values in MESA from slightly pre-MS to slightly post-MS. Endpoints of each evolutionary track are marked with dots of the same colours as the lines.

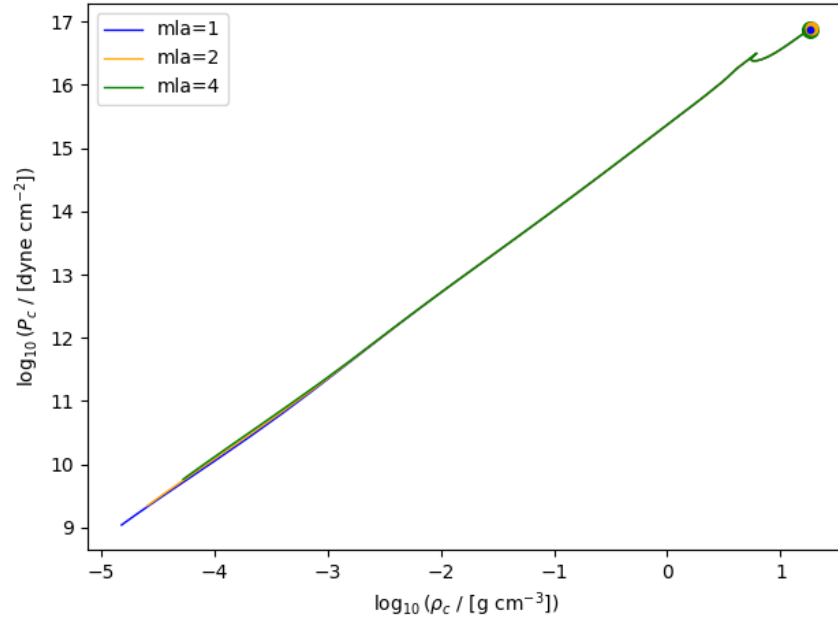


Figure 2: $\log P_c$ vs $\log \rho_c$ plot showing evolution of 15 solar mass stars with varying `mixing_length_alpha` parameter values in MESA from slightly pre-MS to slightly post-MS. Endpoints of each evolutionary track are marked with dots of the same colours as the lines.

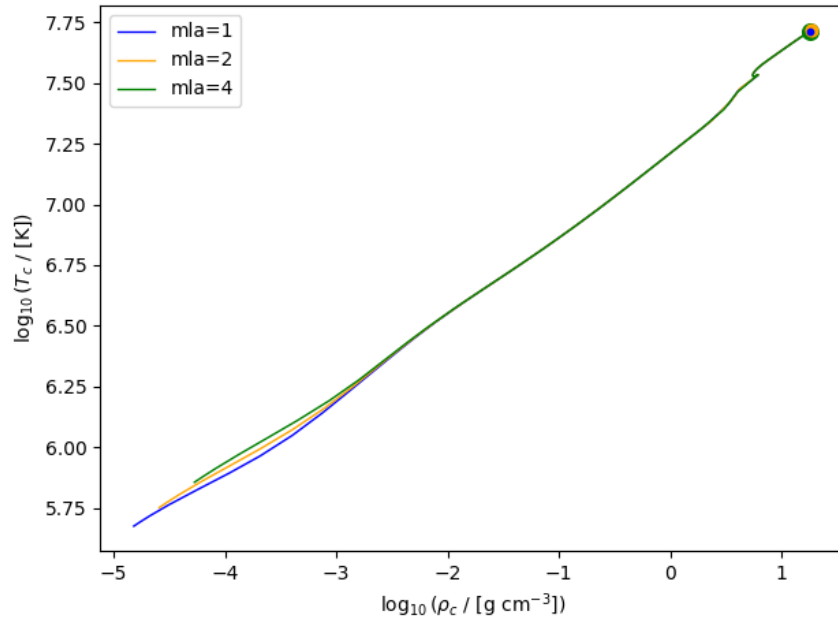


Figure 3: $\log T_c$ vs $\log \rho_c$ plot showing evolution of 15 solar mass stars with varying `mixing_length_alpha` parameter values in MESA from slightly pre-MS to slightly post-MS. Endpoints of each evolutionary track are marked with dots of the same colours as the lines.